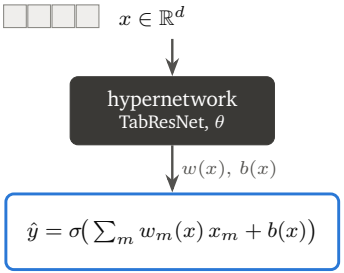
(a) IMN (Kadra et al., 2024)

a black box writes one *linear* model per x



local model = a tangent hyperplane

(b) IMN-TM (this work)

the local model *is* a weighted TM

stage 1 (TM feedback, no gradients)

TM on bits $[x_m \geq t] \Rightarrow$ clauses $c_j(x) \in \{0, 1\}$

frozen after stage 1

stage 2 (back-prop.)

hypernetwork
$x \mapsto W(x)$

$c_1(x) \ldots c_K(x)$

$\hat{y} = \sigma\big(\sum_j W_j(x)\, c_j(x) + b(x)\big)$

explanation: active rules, weighted

+1.4 $x_3 \geq t_1 \wedge \neg\, x_7 \geq t_2$

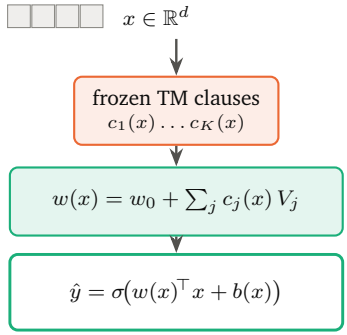
-0.9 $x_1 \geq t_3$

+0.6 $\neg\, x_2 \geq t_4 \wedge x_5 \geq t_5$

local model = constant on a box of x

(c) TM-Gated (this work)

the TM *is* the hypernetwork: no black box



local model = a hyperplane chosen by rules

What we found (12 AutoML-benchmark datasets × 5 seeds, tuned; 60 synthetic tasks)

Better than the TM. IMN-TM beats its own source TM on 12/12 datasets (mean ΔAUROC +0.0094, Holm $p = 0.005$).

Level with the best. Tuned, IMN-TM shares the best average rank with GBDT (3.75; IMN 7.25); vs. IMN: +0.0058, not significant.

More faithful explanations. Neighbour accuracy 0.72 vs. 0.57 for IMN; closer to the true attributions on 41/60 synthetic tasks.

The price is length. 51 (default) to 126 (tuned) weighted rules cover 90% of an IMN-TM explanation, vs. 15 features for IMN.